

MGMT 58600 – Python Programming (Fall 2025)

Class Information

Days: Monday and Wednesday (8 weeks starting from 8/25)

Time: 9:50 – 11:20 pm

Location: RWLS 1057

Contact Information & Office Hours

Instructor: Dr. Shoaib Khan

Email: khan180@purdue.edu

Office Hours: Tuesday and Thursday: 11:30 am – 12:30 pm

TA: TBD

Course Description

In MGMT 586 - Python Programming for business graduate students, we will embark on an exploratory journey into the world of programming with a specific focus on its applications within the realm of business. This eight-week course has been meticulously crafted to empower students with the foundational skills necessary to harness the power of Python for solving real-world business challenges. Through a combination of interactive lessons, hands-on coding exercises, engaging assignments, and collaborative projects, students will gain both theoretical knowledge and practical experience, positioning themselves as adept problem solvers in the modern business landscape.

Course Objectives

1. To provide students with a solid foundation in Python programming.
2. To demonstrate the application of programming in solving business-related tasks.
3. Introduce students to a variety of powerful data processing libraries in Python.
4. To equip students with the ability to analyze business problems and implement effective Python solutions.
5. To encourage collaborative problem-solving and critical thinking through group projects.

Learning Outcomes:

By the end of the course, students should be able to:

1. Demonstrate proficiency in Python programming.
2. Analyze business problems and design Python-based solutions.
3. Develop and implement algorithms to address business challenges.
4. Work collaboratively on coding projects.
5. Apply programming skills to real-world business scenarios.

Technical Requirements

Students will be expected to have uninterrupted access to both computing resources and high-speed Internet connection. In addition, students will be expected to utilize a minimum of a standard desktop or laptop computer (tablets and mobile devices are highly discouraged for the use of this course). The computers must have the most up-to-date drivers and operating system patches/updates.

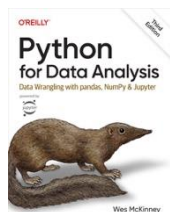
Students must have the ability to install, run, and connect using the Purdue VPN software (Cisco AnyConnect) and also the ability to map an external server drive to their computer of use. This course will require students to save/access files on the Purdue servers (and possibly other locations). Students should also make sure that firewalls on their machine, building, business, service provider, and/or geographical location will not interfere with the ability to run the VPN software or make connections to the servers used in the class (for storing files or running software).

The computing environment should allow for test-taking proctoring/security software to be run without interference (including Respondus). In addition, a microphone and camera must be operational with the computer (for potential class activities, help, and exam settings). Students must also have access to a space where they can guarantee no interruptions during test-taking situations. This should be a room that has sparse accommodations so as not to lead to being flagged as a potential academic misconduct situation (papers or notes hanging on the wall, papers on the desk or around the exam area, etc. should all be avoided). In addition to this, access to the room should be restricted so that other individuals do not enter the room during the exam and so that noises from adjoining rooms do not bleed over into the exam-taking room. Please be specifically careful about clearing papers, technology, or anything else that can be used or construed as being used for academic misconduct from the exam area. Consideration will not be given after-the-fact and there is no need for a proctor to prove that a piece of paper in the exam area (or phone, earbuds, etc.) is being used as it is not supposed to be in the area in the first place.

Learning Resources & Texts

Required Text

Available online free for Purdue students through Purdue library.



[Python for Data Analysis](#)

[Data Wrangling with Pandas, NumPy, & Jupyter](#)

by Wes McKinney

3rd Edition

Released August 2022

Publisher(s): O'Reilly Media, Inc.

ISBN-10: 109810403X

ISBN-13: 978-1098104030

Other Helpful References:

- [Automate The Boring Stuff With Python \(2nd edition\)](#), Al Sweigart, No Starch Press, San Francisco, CA 2015. – The book is available for free.
- <https://www.geeksforgeeks.org/python-programming-examples/>
An excellent resource for revising some python programming concepts

- <https://www.geeksforgeeks.org/python-exercises-practice-questions-and-solutions/>
Very diverse collection of problems and solutions to reinforce python programming concepts
- <https://www.geeksforgeeks.org/data-analysis-with-python/>
Shows examples using snippets of python.
- <https://towardsdatascience.com/a-beginners-guide-to-data-analysis-in-python-188706df5447>
Uses examples to introduce data analysis.

Additional resources that students can check out include:

1. Python 3 (<https://www.python.org/downloads/>)
2. Jupyter (<https://jupyter.org/install>)
3. NumPy (<https://www.numpy.org/>)
4. Pandas (<http://pandas.pydata.org/>)
5. Seaborn (<https://seaborn.pydata.org/>)

Instructor's Online Hours

I will respond to student questions as soon as I am available (generally within 48 hours during the M-F work week). Student inquiries made during the weekend may experience a delayed response time. Questions about the course content, assignments, or lectures should be asked in Piazza. Students are encouraged to answer the questions their peers ask on Piazza. Note that Piazza provides for both private messages as well as public discussion and Q&A postings (both anonymous and non-anonymous), so all course correspondences should occur there.

Assignments (Tentative)

You will have several individual and group assignments throughout the semester.

Your course evaluation will be based on assignments and the exam. The due dates and times for the assignments posted on the course website are in the time date applicable to Purdue University (West Lafayette, IN). Details on these assignments, along with rubrics to guide evaluation, and guidelines on discussion participation and evaluation will be posted on Brightspace.

The composition will be:

Assignments	Responsibility	Points
Assignments (4 - 5)	Individual/Pair	45
Project	Individual/Pair	15
Quizzes (4)	Individual	10
Class Participation	Individual	10
Course Exam	Individual	20
Total		100

Krannert Grading Policy

The target grade distribution for all *elective courses* is 30-40% A/A-, 30-40% B+/B's, 10-20% B-'s, 10-15% C+ or below resulting in approximately an average Grade Point Average (GPA) of 3.35 for each elective course where the GPA is calculated as A = 4, A- = 3.70, B+ = 3.30, B = 3.00, B- = 2.70, C+ = 2.30, C = 2.00.

Participation and Assignment policies

Class Citizenship

Class citizenship means that you come to class each day prepared and ready to learn. Students are expected to attend each class, unless ill. In the case of illness, students should not wait until the next class period to get “caught up” and instead should do so before arriving at the next class period to avoid falling behind. Students are responsible for all material covered in class. If you do not attend and participate in class, that will negatively impact your class participation points. Also, you will likely not perform well on the exams or other graded components of the course. Anecdotally, attendance of all classes and elimination of non-class-related activities during class (such as e-mailing/texting, Facebooking, web surfing, sleeping, doing homework for other classes, etc.) has been shown to have a positive impact on student performance in the course. In some cases, even missing a single class can have a negative impact on student performance in the course which could put a student behind for a week or more. For more information on expected attendance at Purdue, please thoroughly review the website below:

http://www.purdue.edu/studentregulations/regulations_procedures/classes.html

Assignments

There will be up to five individual/pair assignments. Late submissions are accepted, but only up to a maximum of 24 hours beyond the deadline. Anything submitted after the deadline, but before 24 hours beyond the deadline is subject to a 20% late penalty (in addition to any evaluation deductions). Anything submitted after 24 hours beyond the deadline will not be graded and will receive a zero for the assignment. Please note that a minute late is late.

Course Pace

Students are expected to keep up with reading and recording materials. *While it is recognized that some students taking the course have other obligations in addition to the course, students will be expected to treat the course just as any other business obligation and conflicts with workload in other classes or on the job are not grounds for consideration in situations of poor or late performance nor in delay of keeping up with course requirements/expectations.* Students are also expected to treat fellow classmates with the utmost respect in professional/prompt communications, adhering to a Krannert work ethic, and not shirking/free-riding in collaborative lab assignments.

Exam

Students will be expected to take exams at the scheduled time/date. If the exam will be missed because of a University-excused absence (these are typically limited to documented illness, death in the family, required military deployment/training, and some religious observances ... these DO NOT include items like family events/vacations, work conflicts, etc.), you must notify the instructor before the exam in a reasonable fashion commensurate with the circumstances surrounding the absence. When the exam is missed due to a University-excused absence and prior notice is provided, the exam will be rescheduled by the instructor based on the instructor's discretion. The format of the make-up exam and the content may vary from the formats of regularly-scheduled exams. The instructor has the sole right to make determinations concerning the potential for make-up exams. To be safe, never assume that a make-up will be granted; always discuss the situation with your instructor in advance of the absence (with as much time prior to the absences as is reasonably possible). Do not miss an exam, unless you are completely incapacitated in an emergency situation, without first obtaining confirmation from your instructor.

Netiquette

You are encouraged to comment, question, or critique ideas. However, be mindful that sarcasm and humor can be easily misconstrued in online interactions. Please read the Netiquette rules for this course:

- Give other students the opportunity to join in the discussion.
- Present ideas appropriately.
- Be cautious in using Internet language. For example, do not capitalize all letters since this suggests shouting.
- Avoid using vernacular and/or slang language. This could possibly lead to misinterpretation.
- Keep an “open-mind” and be willing to express even your minority opinion.
- Think and edit before you share (e.g., post or email).
- Ask for feedback.

Course Evaluation

During the last two weeks of the course, you will be provided with an opportunity to evaluate this course and your instructor. Purdue now uses an online course evaluation system. You will receive an official email from evaluation administrators with a link to the online evaluation site. You will have up to two weeks to complete this evaluation. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the evaluation system.

Academic Dishonesty

It is your responsibility to understand and abide by the Purdue University Student Code of Conduct. Lab assignments may be completed with a single partner unless otherwise stated. Exams are to be individual work, are closed book, closed note, and are not collaborative. No electronics are to be used or held or accessed while taking an exam, including cell phones, pagers, pdas, or other similar devices. Please see the following Purdue websites for details about definitions, university procedures for, and consequences for violations of academic integrity:

- https://www.purdue.edu/odos/osrr/resources/documents/academic_integrity.html
- https://www.purdue.edu/odos/osrr/resources/documents/responding_to_academic_dishonesty.html
- http://www.purdue.edu/studentregulations/student_conduct/regulations.html

These websites are meant to be a guide and not an all-inclusive list of academic misconduct activities. The spirit of the law, in addition to the letter of the law, will always be used.

The use of another student's or entity's work (outside of your one specified partner on a given assignment), in part or in whole (especially programming code), with or without the other student's/entity's permission, will be considered grounds for academic misconduct prosecution. There will be many opportunities during office hours or other interactions with your instructor to obtain help if you do not understand the concepts.

Any good management student should understand cost/benefit analysis and any benefit that you might perceive about academic misconduct is far exceeded by its cost. Instructors for this course have been instructed to vigorously investigate and prosecute any cases of academic misconduct. Note that, in most cases, the penalties given by the instructor may be JUST THE BEGINNING. The Krannert School of Management and Purdue University, through the Dean of Students office, reserve the right to add penalties BEYOND those imposed by the instructor. These include failure of the course, suspension from the University or school, and potentially expulsion from the University or school. Lastly, do not play games with the rationalization of the scope of academic misconduct. Even small infractions may lead to very big consequences up to and including failure of the course, suspension from the University or School, and expulsion from the University or School.

Turnitin or similar software may be used as needed to detect plagiarism.

Students who observe or suspect academic or conduct violations are encouraged to get in touch with their instructor immediately and/or report the issue directly to the Dean of Students Office via the following online link: <https://publicdocs.maxient.com/incidentreport.php?PurdueUniv>.

Extra Credit and Deadline Extensions:

In the business world, programming component deadlines are extremely important and thus we will treat assignment deadlines as we do in the real world. No assignment deadline extensions or extra credit will be provided. Quizzes, assignments, and examinations will be considered for re-grading ONLY if turned into the instructor within 5 business days of being returned and accompanied by a written explanation of the problem. No verbal explanations are accepted. The instructor reserves the right, in such cases, to re-grade the entire exam, assignment, or quiz. In the case of final exams, students should submit the request for re-grading no later than the end of the first week of the spring or fall semester following the semester in which the grade was assigned.

As mentioned before, students are expected to take every exam at the scheduled time/date. In a case where serious emergency prevents a student from doing so, the instructor must be notified immediately. Again, students must notify the instructor IN ADVANCE of any missed exam unless there is an emergency situation (such as having to go to the hospital by ambulance or getting in a car wreck on the way to the exam). All emergencies and absences must be fully documented by the appropriate authority (e.g., the health-care provider concerned in the case of a medical emergency). If you miss a class because of an illness, you must see the doctor no later than the day of the absence. If you see the doctor before the day of the missed class period, your documentation from that visit must include some indication from the doctor that the illness or treatment is sufficient to keep you out of class beyond the day of the visit. Visits to the doctor after the day of the exam/presentation will be an indication that the "illness" was not severe enough to warrant missing the exam and thus will not be considered as valid supporting documentation.

All materials provided for the course, including lectures, videos, articles, etc. are the protected property of the instructor and/or other third parties and are protected by state, federal, and international law. Students may not record, stream, etc. any video/audio of lectures or part of a lecture (no matter how small) without the written consent of the instructor. In addition, no provided materials may be transferred, streamed, uploaded, etc. by any means to any location without written consent of the instructor (this includes, but is not limited to social media sites, student "study" sites, etc.). Under no circumstances is any material obtained from the instructor in this course to be used for personal or business gain without written consent of the instructor or the material's owner.

Emergency Statement

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

Disability Statement

Any student in this course who has a disability that may prevent him or her from fully demonstrating his or her abilities should contact the instructor, as soon as possible, so DRC-approved accommodations necessary to ensure full participation and facilitate your educational opportunities can be discussed. In accordance with University guidelines, students requesting to activate an accommodation must do so no later than the end of the third class session of the semester which is less than 16 weeks. As this course does not specifically have "class periods," students should notify the instructor no later than 5:00 pm on Friday of the first week of the

semester/module. An activation may always be discontinued, but it cannot be activated after this time frame.

Students who are having problems with test taking (or any activities) in this class are strongly encouraged to contact their instructor for help as soon as there is a problem to obtain the necessary guidance.

Nondiscrimination

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. Purdue University prohibits discrimination against any member of the University community on the basis of race, religion, color, sex, age, national origin or ancestry, genetic information, marital status, parental status, sexual orientation, gender identity and expression, disability, or status as a veteran. Any student who believes they have been discriminated against may visit www.purdue.edu/report-hate to submit a complaint to the Office of Institutional Equity. Information may be reported anonymously.

Academic Guidance in the Event a Student is Quarantined/Isolated

If you become quarantined or isolated at any point in time during the semester you will have access to an Academic Case Manager who can provide you academic support. Your Academic Case Manager can be reached at acmq@purdue.edu and will provide you with general guidelines/resources around communicating with your instructors, be available for academic support, and offer suggestions for how to be successful when learning remotely. Importantly, if you find yourself too sick to progress in the course, notify your academic case manager and notify me via email or Brightspace. We will make arrangements based on your particular situation. The Office of the Dean of Students (odos@purdue.edu) is also available to support you should this situation occur.

Course Schedule (Tentative)

Class	Date	Class Topic and/or Reading	Assignments Due
			Assignment Due (Monday of the Week) 11:59 pm EST on Brightspace
1	8/26	Introduction to python, a quick recap of code basics including variables, basic data types and operators.	
2	8/28	Conditional statements and Loops	
3	9/2	Basic data structures in python – strings, lists, dictionaries, tuples. Introduction to loops.	
4	9/4	Case Study: Applying tools of programming to solve a business problem	
			Assignment # 1 Due (11:59 pm EST 9/8)
5	9/9	Writing functions, modules, handling errors and exceptions.	
6	9/11	File handling in Python – read/write from/to files.	
			Assignment # 2 Due (11:59 pm EST 9/15)
7	9/16	Introduction to NumPy	
8	9/18	Introduction to Pandas	
			Assignment # 3 Due (11:59 pm EST 9/22)
9	9/23	File I/O with Pandas, working with dataframes. Introduction to descriptive statistics.	
10	9/25	Intro to plotting with Pandas, Matplotlib and Seaborn. Data visualizations - 1D (box plots, histograms, bar plots) and 2D (scatter plots, pair plots, joint plots).	
			Assignment # 4 Due (11:59 pm EST 9/29)
11	9/30	Web Scraping using Python.	
12	10/2	Advanced Python: Lambda functions, Filter and Map Functions.	
			Assignment # 5 Due (11:59 pm EST 10/6)
13	10/7	Vibe Coding: Tools and Techniques	
14	10/9	Brief overview of Python's predictive analytics tools and libraries.	
			Project Due (11:59 pm EST 10/13)
15	TBA	Final Exam	